

CNS

By Dr.Riffat Ibrahim Riffat

General Anesthetics



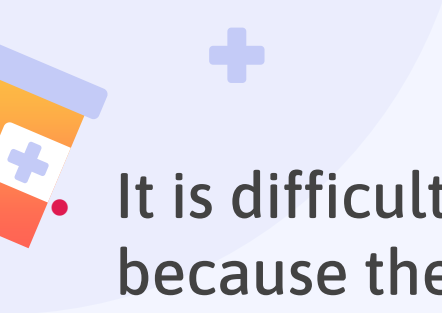
Introduction

General anesthesia is intended to produce 5 desired effects:

- Unconsciousness
- amnesia
- Analgesia
- skeletal muscle relaxation
- and inhibition of autonomic reflexes

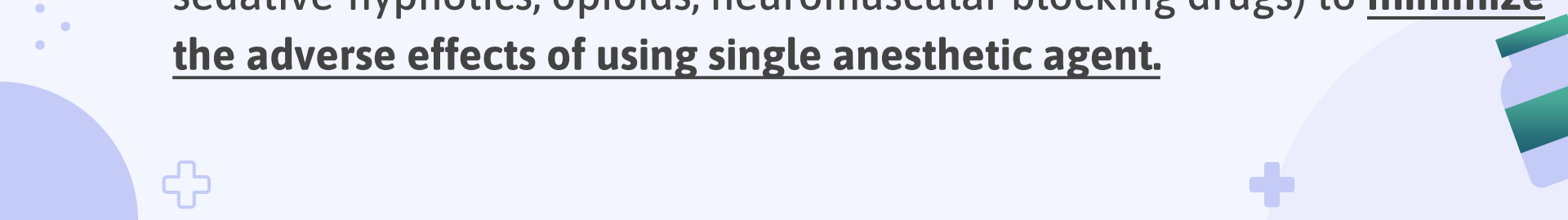


- The purpose of these effects is loss of response to surgical incision.



• It is difficult to obtain these 5 desired effects with a single drug because the concentration needed to achieve all these effects will be toxic. So, it is better to use more than one drug to achieve them without the need to increase the dose (balanced anesthesia).

• **Balanced anesthesia:**

- Is employing multiple drugs (IV anesthetics, inhaled anesthetics, sedative-hypnotics, opioids, neuromuscular blocking drugs) to minimize the adverse effects of using single anesthetic agent.
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• Phases of anesthesia:

• 1. Before surgery (preanesthetic medications):

- One hour before operation, the patient is given anxiolytic to relieve anxiety and may be given metoclopramide to hasten gastric evacuation and prevent vomiting and aspiration pneumonia.

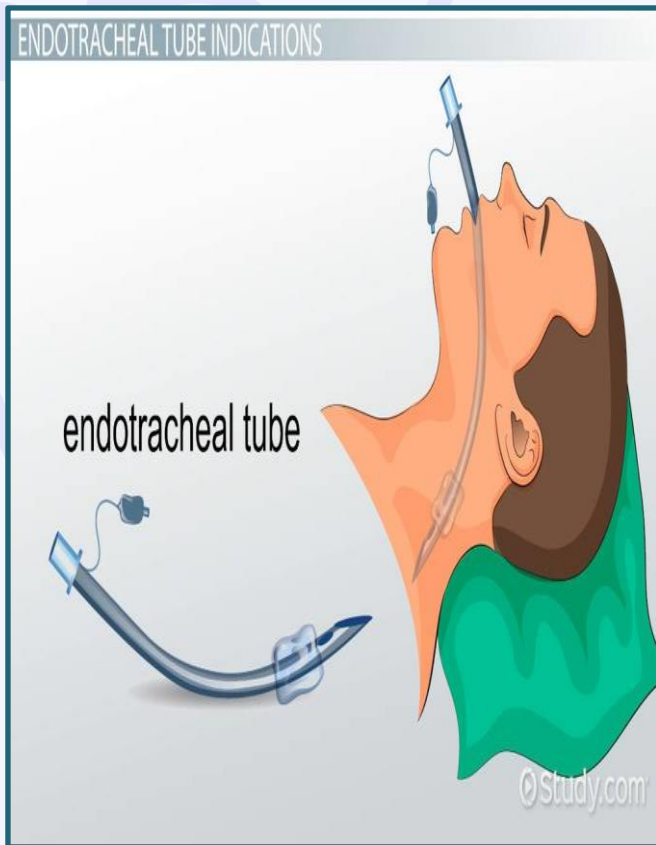
• 2. During surgery:

- Induction of anesthesia usually by an IV anesthetic
- Maintenance is achieved by inhalation anesthetic
- The patient is also given NMBs

• 3. After surgery:

- The patient is not left until respiration is normal
- Relieve of pain by an opioid





Classification of general anesthetics:

- I. Inhalation agents:
 - Volatile liquids: isoflurane – sevoflurane – desflurane – enflurane – halothane (halogenated anesthetics)
 - Gases: nitrous oxide – xenon
- II. IV agents:
 - Propofol – thiopental – etomidate – ketamine – midazolam



Pharmacodynamics

- 1. Activation of GABAA receptor:
- Most anesthetic agents directly and indirectly facilitate a GABA-mediated increase in chloride conductance to hyperpolarize and inhibit neuronal membrane activity.
- Examples:
 - Propofol, etomidate, isoflurane, benzodiazepines (midazolam, lorazepam, diazepam), and barbiturates (e.g. sodium thiopental)

Pharmacodynamics

- 2. Blocking of NMDA receptor: Nitrous oxide and ketamine inhibit excitatory glutamate-gated ion channels (NMDA receptors).
- 3. Opening of two-pore potassium channels (K2P): Opening of these channels causes neuronal hyperpolarization which reduces neuronal excitability.
 - Examples: halogenated inhalation anesthetics

Adverse effects of inhaled anesthetics:

Malignant hyperthermia:

Is a rare (1:20000), but life-threatening condition that may occur when anesthetics are used with NMBs like succinylcholine.

It is due to genetic abnormality in Ca^{2+} release channels (ryanodine receptor) in sarcoplasmic reticulum (SR) of muscle cells, leading to massive release of Ca^{2+} from SR following exposure to inhaled anesthetics or succinylcholine.

- **Symptoms include:** muscle spasm, hyperthermia, hypertension, tachycardia, and hyperkalemia.
- **Treatment includes:** dantrolene, a muscle relaxant that blocks Ca^{2+} release channels + supportive measures.

Adverse effects of inhaled anesthetics:

2. CNS: Rise of IC pressure due to VD of cerebral blood vessels.

3. CVS:

- Hypotension: due to myocardial depression and peripheral VD.
- Cardiac arrhythmia: especially with halothane.

4. Respiratory:

- Desflurane is a pulmonary irritant and can cause bronchospasm.
- Inhibition of mucociliary function.

5. Liver: Halothane can cause acute liver injury (halothane hepatitis).

6. Uterus: Inhaled anesthetics cause uterine relaxation which can be helpful when uterine relaxation is required for intrauterine fetal manipulation or manual extraction of a retained placenta. However, it can also lead to increased uterine bleeding after delivery when uterine contraction is desired.



+ INTRAVENOUS ANESTHETICS



- Propofol – thiopental – etomidate – ketamine – midazolam
 - They are commonly used for rapid induction of anesthesia.
 - IV anesthetics are highly lipophilic drugs, upon administration, they rapidly distribute in the highly vascular tissues including the brain causing rapid induction of anesthesia.
 - Termination of anesthesia is due to redistribution of the drug from nervous tissue to other tissues such as muscles, viscera, and adipose tissue.
 - Prolonged administration (cont. IV infusion) can lead to saturation of adipose tissue and slowing of the redistribution phase → prolongation of anesthesia, however, this can lead to undesired side effects.
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Propofol (Diprivan 1%)

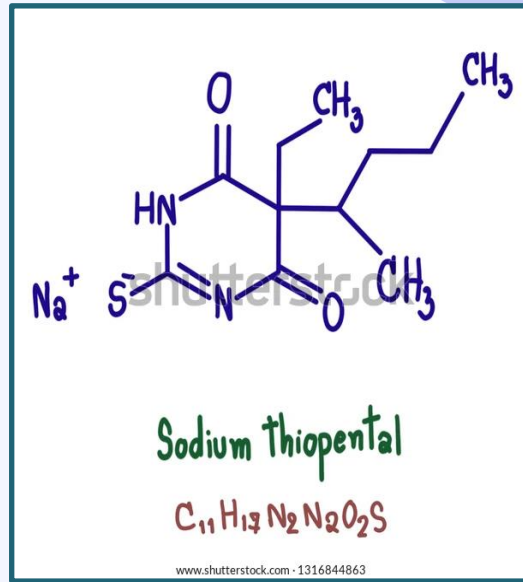
The most commonly used IV agent for induction of anesthesia.

- It is administered as an oil-in-water emulsion which can cause pain at the injection site. **Fospropofol** is a water-soluble derivative that is less painful and rapidly converted to propofol in the body.
- Adverse effects:
 - Pain at the site of injection
 - Propofol infusion syndrome occurs with prolonged infusion and consists of severe metabolic acidosis, skeletal muscle necrosis (rhabdomyolysis), hyperkalaemia, lipaemia, hepatomegaly, renal failure, arrhythmia and cardiovascular collapse.



Thiopental

- Thiopental is the only remaining barbiturate in common use.
- Used only for induction of anesthesia (not for maintenance).
- It is highly lipophilic drug. On IV injection, thiopental causes unconsciousness within about 20s, lasting for 5–15 min before it redistributes to fatty tissues.
- Adverse effects:
 - High dose can cause hypotension, RC depression, and myocardial depression.
 - The solution of thiopental is highly alkaline, if injected SC or IM, it causes local tissue necrosis and ulceration that can result in gangrene.
 - Like other barbiturates, it is hepatic microsomal enzyme inducer.



Ketamine (Ketalar)

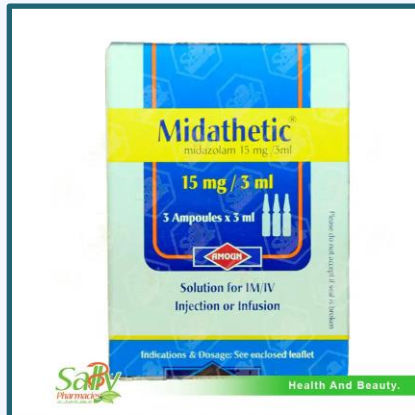
The characteristic state observed after an induction dose of ketamine is known as “dissociative anesthesia,” wherein the patient’s eyes remain open with a slow nystagmic gaze.

- Ketamine’s mechanism of action includes inhibition of the NMDA receptor.
- It differs from most other IV anesthetics in:
 - It produces significant analgesia
 - It raises the blood pressure
 - Does not affect respiration.
- The adverse effects are less marked in children making it preferred in pediatrics.
 - Increase IC tension
 - Hallucinations and psychiatric disturbances on recovery.



Midazolam (Dormicum)

- It is a very short acting BZ. It has rapid onset and short duration.
- May be used preoperatively for sedation and to reduce anxiety.
- It is used as a sole agent for surgical and diagnostic procedures that do not require analgesia (endoscopy, cardiac catheterization).
- The actions of the BZs can be reversed with flumazenil.



Cases

- Which of the following is correct regarding malignant hyperthermia?
- A. It is triggered by dantrolene.
- B. It is triggered by local anesthetics.
- C. It is generally mild and clinically insignificant. D. It has no familial component.
- E. It involves increased skeletal muscle oxidative metabolism.

Answers

- Correct answer = E. Malignant hyperthermia involves increased skeletal muscle oxidative metabolism and is a life-threatening condition. Dantrolene is the specific pharmacologic treatment. Local anesthetics have been shown to be safe. Triggering agents include succinylcholine and halogenated hydrocarbon volatile anesthetic agents in susceptible individuals. Susceptibility to malignant hyperthermia is inherited in an autosomal dominant fashion.

Cases

- Which of the following agents is preferred I.V agent for children
 - A. Ketamine
 - B. Propofol
 - C. isoflurane
 - D. NO

Answers

- Correct answer = A, The adverse effects are less marked in children making it preferred in pediatrics.

Cases

- A 36-year-old man I said to have an endoscopy, which of the following agents can be used as the sole anesthetic for this procedure
 - A. Ketamine
 - B. Propofol
 - C. Midazolam
 - D. sevofulrane

Answers

- Correct answer = D. Midazolam is used as a sole agent for surgical and diagnostic procedures that do not require analgesia (endoscopy, cardiac catheterization).